

chased out of Europe by the Nazis. Wald was a senior member of the SRG, and his advice to the military: “Put the armor where the bullet holes aren’t” had a huge impact on the war.

It’s counterintuitive, but it makes sense when you think about it: Shot-up planes that returned meant the areas they were hit were not critical to the plane’s survivability. Wald’s great insight—“Where were the holes on the bombers that *weren’t* coming back from their missions?”—was crucial to understanding the problem.

Looking only at where returning aircraft had been shot has become one of history’s most famous example of **survivorship bias**. Wald taught us to always consider the full data set—including set members *you cannot see*.

This tendency to give too much weight to the survivors while excluding the deceased has a long and storied history in the mutual fund industry. Investors first learned what survivorship bias was in the 1990s, after research revealed that fund returns were being significantly overstated.¹³⁹ Ignore the funds that closed due to poor performance, leave in only the surviving winners, and the data will look a whole lot better! Add back those dead or retired funds (otherwise removed from the dataset), and the touted investment outperformance disappeared.

Investing giant Vanguard Group found that 62% of surviving large-capitalization value funds outperformed their specific benchmark. But that was ignoring funds that were closed. Adding them back into the dataset lowers the performance metric to less than half—just 46% after five years.

Dimensional Funds Advisors (DFA) found about half of all mutual funds close within 15 years. The non-surviving funds—closed, merged into a more successful fund, otherwise retired—were usually killed off due to poor performance. DFA’s analysis found that by only looking at the surviving strategies, the data suffers from a substantial upward bias that overstates the median fund outperformance (alpha) by 50%.¹⁴⁰ Killing off